

Computational and AI Methods in Climate Economics

Lecture 2 (Part 2) — Climate meets the economy: the DICE/CDICE model, solved with Deep Equilibrium Nets

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Based on: Folini, Friedl, Kübler & Scheidegger (2025), “The Climate in Climate Economics,”
Review of Economic Studies 92(1), 299–338, and its Online Appendix.

Production code: https://github.com/ClimateChangeEcon/Climate_in_Climate_Economics

Materials: https://github.com/sischi/cemracs2026_DEQN

Roadmap of This Lecture

Part A

- ▶ **Part I.** Motivation: why climate economics, why IAMs, why this is computationally hard
- ▶ **Part II.** The DICE/CDICE model: economy, carbon cycle, temperature, damages, planner problem, social cost of carbon

Part B

- ▶ **Part III.** Solving DICE with DEQNs: time as a state, detrending, shadow prices as network outputs, the 8-component loss, training, accuracy
- ▶ **Part IV.** What is different compared to the Lecture 1 models (Brock–Mirman, IRBC)?

Learning objectives

By the end of this lecture you should be able to:

- ▶ Describe the building blocks of DICE/CDICE: neoclassical growth economy, 3-reservoir carbon cycle, 2-layer temperature dynamics, damages, and abatement
- ▶ Explain why an IAM is **non-stationary** and what that implies for recursive solution methods
- ▶ Map a non-stationary model onto DEQNs: time as a state variable, the Traeger time transformation, detrending to efficiency units
- ▶ Derive the 8-component DEQN loss from the planner's Lagrangian, including the Fischer–Burmeister treatment of $0 \leq \mu_t \leq 1$
- ▶ Read the social cost of carbon directly off the learned shadow prices, and contrast the whole recipe with the Lecture 1 models

Part I

Why Climate Economics?

Climate Change as an Economic Threat



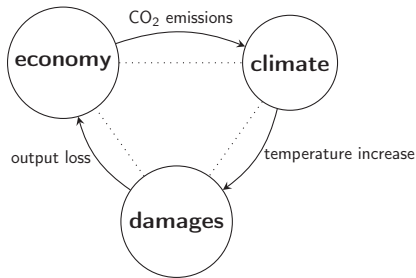
Observed global warming motivates the IAM feedback loop below.

- ▶ Global warming is a growing threat to economic well-being.
- ▶ Tipping points place us in catastrophic danger.
- ▶ Impacts current and future generations in different regions very differently.
- ▶ Recent overviews: Dietz (2024), van der Ploeg and Rezai (2026), Fernandez-Villaverde, Gillingham, Scheidegger (2025).

Why Integrated Assessment Models (IAMs)?

IAM logic

Couple the **economy** (output, investment, abatement, welfare) with the **climate system** (carbon concentrations, temperature, damages).



Output of interest: social cost of carbon (SCC), optimal tax paths, welfare decomposition.

The Social Cost of Carbon (SCC)

Definition

SCC at time t : marginal welfare cost of one extra unit of emissions, converted to consumption units.

$$SCC_t = -\frac{\partial V_t / \partial E_t}{\partial V_t / \partial C_t}$$

High SCC when

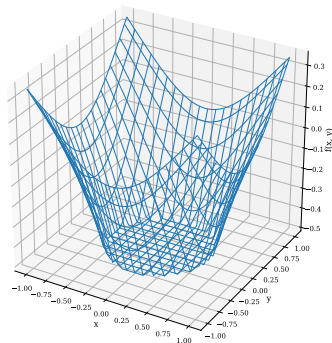
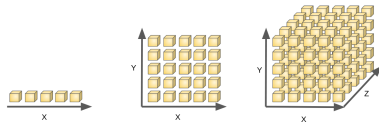
- ▶ Damages are steep
- ▶ Climate response is strong
- ▶ Discounting is low
- ▶ Tipping risk is material

Policy interpretation

- ▶ Tax path for carbon pricing
- ▶ Benchmark for regulation / cap-and-trade
- ▶ Input to cost-benefit analysis
- ▶ Why uncertainty matters: SCC is a **distribution**

Why This Is Computationally Hard

- ▶ **Nonstationarity:** TFP, population, emissions intensity, and forcing paths all change exogenously over time.
- ▶ **Coupled dynamics:** economy and climate interact in both directions.
- ▶ **Long horizon:** welfare effects unfold over 100–300 years.
- ▶ **Curse of dimensionality:** nonlinearities, tipping hazards, constraints.



This Lecture's Approach

- ▶ **Take the DEQN machinery from Lecture 1** — policy networks trained on the residuals of equilibrium conditions along simulated paths — and apply it to a **non-stationary** climate–economy model.
 - ▶ **Solve the deterministic CDICE benchmark globally**, so that the social cost of carbon comes out as a by-product of the learned shadow prices.
1. State the **DICE/CDICE model**: economy + carbon cycle + temperature + damages.
 2. Map it onto **deep equilibrium nets**: time as a state, detrending, multipliers as outputs.
 3. Train on the **8 first-order conditions** and validate with error statistics.

Connection to Part 1 of this lecture: once the model can be solved fast and reliably, GP surrogates and Bayesian active learning turn the SCC into a *distribution* over uncertain parameters.

Part II

The DICE/CDICE Model

DICE: A Minimal Global IAM

State variables

- ▶ Economic stock: K_t
- ▶ Carbon stocks: $M_t^{\text{AT}}, M_t^{\text{UO}}, M_t^{\text{LO}}$
- ▶ Temperature states: $T_t^{\text{AT}}, T_t^{\text{OC}}$
- ▶ Exogenous trends: population L_t , TFP A_t , carbon intensity σ_t

Controls

- ▶ Consumption C_t (equivalently investment I_t)
- ▶ Abatement rate $\mu_t \in [0, 1]$

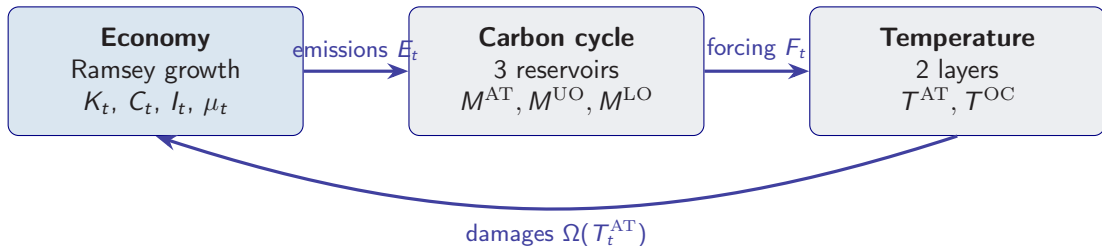
Why DICE is influential

- ▶ Transparent
- ▶ Fast to evaluate
- ▶ Policy-facing
- ▶ Benchmark for extensions

Challenge

Fast is useful, but only if the climate module is calibrated consistently with climate science — this is where **CDICE** comes in.

Anatomy: One Economy, Two Physical Blocks



- ▶ The carbon cycle affects temperature, **but not vice versa**: the physical part is a one-directional chain, closed only through the damage function.
- ▶ All coefficients carry an explicit time step Δ_t : DICE-2007 uses $\Delta_t = 10$ years, DICE-2016 uses 5, CDICE uses $\Delta_t = 1$ — one **generic formulation** covers all of them.

The Economy I: Production and Capital

Standard neoclassical growth, with output produced from capital and effective labor:

$$Y_t^{\text{Gross}} = (A_t L_t)^{1-\alpha} K_t^\alpha,$$
$$K_{t+1} = (1 - \delta^K) \Delta_t K_t + \Delta_t I_t.$$

Ingredients

- ▶ A_t : total factor productivity (exogenous trend)
- ▶ L_t : population = labor (exogenous trend)
- ▶ $\alpha = 0.3$, $\delta^K = 0.1$ per year

So far, so familiar

Without the climate block this is the deterministic growth model from Lecture 1 — **except** that A_t and L_t follow exogenous time paths, so the model has no steady state to converge to.

The Economy II: Damages and Abatement

Warming destroys a fraction of gross output; abating emissions costs another fraction:

$$\Omega_t = \psi_1 T_t^{\text{AT}} + \psi_2 (T_t^{\text{AT}})^2$$

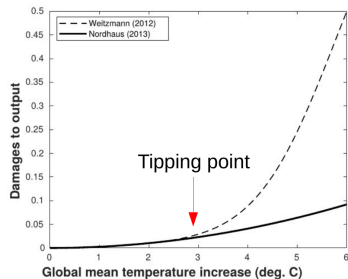
(damages, DICE-2016: $\psi_1 = 0$, $\psi_2 = 0.00236$)

$$\Theta_t = \theta_{1,t} \mu_t^{\theta_2}$$

(abatement cost, $\theta_2 = 2.6$, $\theta_{1,t}$ falling backstop price)

$$Y_t = Y_t^{\text{Gross}} (1 - \Theta_t - \Omega_t), \quad C_t = Y_t - I_t.$$

See also Tol (2009), Hänsel et al. (2020)



- $\mu_t \in [0, 1]$: fraction of industrial emissions abated; $\mu_t = 1$ means the **backstop technology** eliminates all emissions.
- The degree of convexity of Ω is of key importance for optimal taxes (Weitzman 2012); tipping variants are much steeper.
- DICE-2007 uses the hyperbolic form $\Omega_t = 1/(1 + \psi_1 T_t^{\text{AT}} + \psi_2 (T_t^{\text{AT}})^2)$ — same generic structure.

Damage fraction $\Omega(T)$: share of gross output lost at warming T .

The Economy III: Preferences

A social planner maximizes population-weighted CRRA utility of per-capita consumption:

$$U = \sum_{t=0}^T \beta^t \Delta_t \frac{\left(\frac{C_t}{L_t}\right)^{1-1/\psi} - 1}{1 - 1/\psi} L_t, \quad \beta = \frac{1}{(1 + \rho)^{\Delta_t}}.$$

Parameters (DICE-2016/CDICE)

- ▶ IES $\psi = 0.69$
- ▶ Rate of time preference $\rho = 0.015$ per year
- ▶ Initial capital $K_0 = 223$ trillion USD (2015)

Discounting is the battleground

With effects unfolding over centuries, ρ and ψ dominate the size of the SCC — one reason global sensitivity analysis (Part 1 of this lecture) matters so much in climate economics.

The Climate I: Carbon Cycle (3-Box Model)

Three carbon reservoirs — atmosphere (AT), upper ocean (UO), lower ocean (LO) — exchange carbon as a **linear dynamical system**, $M_{t+1} = (I + \Delta_t B) M_t + \Delta_t E_t$:

$$M_{t+1}^{\text{AT}} = (1 - \Delta_t b_{12}) M_t^{\text{AT}} + \Delta_t b_{12} \frac{M_{\text{EQ}}^{\text{AT}}}{M_{\text{EQ}}^{\text{UO}}} M_t^{\text{UO}} + \Delta_t E_t,$$

$$M_{t+1}^{\text{UO}} = \Delta_t b_{12} M_t^{\text{AT}} + (1 - \Delta_t b_{12} \frac{M_{\text{EQ}}^{\text{AT}}}{M_{\text{EQ}}^{\text{UO}}} - \Delta_t b_{23}) M_t^{\text{UO}} + \Delta_t b_{23} \frac{M_{\text{EQ}}^{\text{UO}}}{M_{\text{EQ}}^{\text{LO}}} M_t^{\text{LO}},$$

$$M_{t+1}^{\text{LO}} = \Delta_t b_{23} M_t^{\text{UO}} + (1 - \Delta_t b_{23} \frac{M_{\text{EQ}}^{\text{UO}}}{M_{\text{EQ}}^{\text{LO}}}) M_t^{\text{LO}}.$$

Emissions couple the economy into the carbon cycle:

$$E_t = \sigma_t Y_t^{\text{Gross}} (1 - \mu_t) + E_t^{\text{Land}}$$

- Mass conservation: columns of B sum to zero; no direct $\text{AT} \leftrightarrow \text{LO}$ exchange.
- Free parameters: transfer rates b_{12} , b_{23} and equilibrium ratios $M_{\text{EQ}}^{\text{AT}}/M_{\text{EQ}}^{\text{UO}}$, $M_{\text{EQ}}^{\text{UO}}/M_{\text{EQ}}^{\text{LO}}$.

The Climate II: Temperature (2-Layer Energy Balance)

Atmosphere-plus-upper-ocean layer T^{AT} and deep-ocean layer T^{OC} :

$$\begin{aligned}T_{t+1}^{\text{AT}} &= T_t^{\text{AT}} + \Delta_t c_1 F_t - \Delta_t c_1 \frac{F_{2\text{XCO}_2}}{T_{2\text{XCO}_2}} T_t^{\text{AT}} - \Delta_t c_1 c_3 (T_t^{\text{AT}} - T_t^{\text{OC}}), \\T_{t+1}^{\text{OC}} &= T_t^{\text{OC}} + \Delta_t c_4 (T_t^{\text{AT}} - T_t^{\text{OC}}),\end{aligned}$$

driven by radiative forcing from atmospheric carbon (plus exogenous non- CO_2 forcing):

$$F_t = F_{2\text{XCO}_2} \frac{\log(M_t^{\text{AT}} / M_{\text{base}}^{\text{AT}})}{\log(2)} + F_t^{\text{EX}}.$$

Key parameter: equilibrium climate sensitivity (ECS)

$T_{2\text{XCO}_2}$ is the long-run warming from a doubling of atmospheric CO_2 ; the feedback strength is $\lambda = F_{2\text{XCO}_2} / T_{2\text{XCO}_2}$. Physically, $1/c_1$ and c_3/c_4 are heat capacities, c_3 the heat-exchange coefficient.

Exogenous Trends: The Source of Non-Stationarity

Six deterministic, time-dependent processes drive the model (DICE-2016 forms, annual t):

Population: $L_t = L_0 + (L_\infty - L_0) \left(1 - e^{-\Delta_t \delta^L t}\right)$ growth g_t^L

TFP: $A_t = A_0 \exp\left(\frac{\Delta_t g_0^A (1 - e^{-\Delta_t \delta^A t})}{\Delta_t \delta^A}\right)$ growth $g_t^A = \Delta_t g_0^A e^{-\Delta_t \delta^A t}$

Carbon intensity: $\sigma_t = \sigma_0 \exp\left(\frac{\Delta_t g_0^\sigma}{\log(1 + \Delta_t \delta^\sigma)} ((1 + \Delta_t \delta^\sigma)^t - 1)\right)$

Backstop cost: $\theta_{1,t} = \frac{p_0^{\text{back}} e^{-g^{\text{back}} t} 1000 \cdot c2co2 \cdot \sigma_t}{\theta_2}$

Land emissions: $E_{\text{Land},t} = E_{\text{Land},0} e^{-\Delta_t \delta^{\text{Land}} t}$

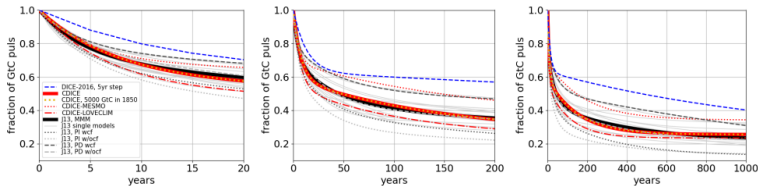
Non-CO₂ forcing: $F_t^{\text{EX}} = F_0^{\text{EX}} + \frac{1}{T/\Delta_t} (F_1^{\text{EX}} - F_0^{\text{EX}}) \min(t, T/\Delta_t)$

Why this matters for us

Because $A_t, L_t, \sigma_t, \theta_{1,t}, E_{\text{Land},t}, F_t^{\text{EX}}$ all depend on calendar time, the optimal policy is **not time-invariant**: $C_t \neq C(K_t, M_t, T_t)$ alone. This single fact drives most of the DEQN design choices in Part III.

From DICE to CDICE: Why Recalibrate?

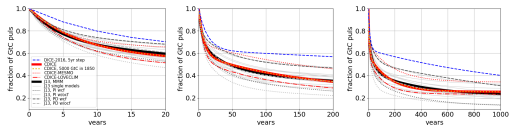
- ▶ Climate emulators must reproduce benchmark behavior from climate-science model archives (CMIP5).
- ▶ Earlier critiques highlighted major discrepancies for common IAM calibrations: DICE-2016's carbon cycle and temperature blocks are **not in balance** and overreact to changes in inputs.
- ▶ Folini et al. (2025) show the **functional form can remain simple**, but the parameters require systematic calibration — the result is **CDICE**.



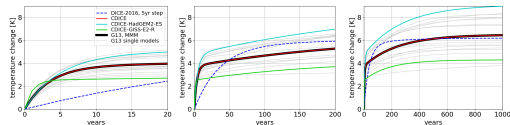
CDICE recalibrates the reduced-form climate emulator to climate-science benchmarks, on an annual time step ($\Delta_t = 1$).

CDICE Calibration: Benchmarks from Climate Science

Test	Targets	Use
1. Carbon pulse (100 GtC)	Atmospheric retention path	Calibrate carbon cycle
2. $4\times\text{CO}_2$ step	Temperature impulse response	Calibrate temperature block
3. 1% CO_2/year	Transient climate response (TCR)	Out-of-sample validation
4. Historical + RCP	Realistic forcing paths	End-to-end validation



Carbon-pulse response.



Temperature response to a $4\times\text{CO}_2$ forcing step.

CDICE at a Glance: Key Parameter Values

Economy (annual)

	Symbol	CDICE
Capital depreciation	δ^K	0.1
Capital share	α	0.3
Damage (quadratic)	ψ_2	0.00236
Abatement exponent	θ_2	2.6
IES	ψ	0.69
Time preference	ρ	0.015
Initial capital [tn USD]	K_0	223

Climate (annual)

	Symbol	CDICE
Carbon transfer AT→UO	b_{12}	0.054
Carbon transfer UO→LO	b_{23}	0.0082
Equilibrium masses [GtC]	M_{EQ}	(607, 489, 1)
Initial masses 2015 [GtC]	M_{INI}	(851, 628, 1)
Temperature coefficients	c_1, c_3, c_4	0.137, 0.73, 0.
Forcing CO ₂ doubling	$F_{2\times CO_2}$	3.45 W/m ²
ECS	$T_{2\times CO_2}$	3.25 °C
Initial temperatures	T_0	(1.1, 0.27)

Full generic calibration tables (DICE-2007, DICE-2016, CDICE side by side): Online Appendix of Folini et al. (2025) — in your readings/ folder.

The Planner's Problem (Annual CDICE, $\Delta_t = 1$)

$$\max_{\{C_t, \mu_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \frac{\left(\frac{C_t}{L_t}\right)^{1-1/\psi} - 1}{1 - 1/\psi} L_t$$

$$\text{s.t. } K_{t+1} = (1 - \Omega(T_{\text{AT},t}) - \Theta(\mu_t)) K_t^\alpha (A_t L_t)^{1-\alpha} + (1 - \delta) K_t - C_t \quad (\lambda_t)$$

$$M_{t+1}^{\text{AT}} = (1 - b_{12}) M_t^{\text{AT}} + b_{12} \frac{M_{\text{EQ}}^{\text{AT}}}{M_{\text{EQ}}^{\text{UO}}} M_t^{\text{UO}} + \sigma_t (1 - \mu_t) K_t^\alpha (A_t L_t)^{1-\alpha} + E_t^{\text{Land}} \quad (\nu_t^{\text{AT}})$$

$$M_{t+1}^{\text{UO}} = b_{12} M_t^{\text{AT}} + \left(1 - b_{12} \frac{M_{\text{EQ}}^{\text{AT}}}{M_{\text{EQ}}^{\text{UO}}} - b_{23}\right) M_t^{\text{UO}} + b_{23} \frac{M_{\text{EQ}}^{\text{UO}}}{M_{\text{EQ}}^{\text{LO}}} M_t^{\text{LO}} \quad (\nu_t^{\text{UO}})$$

$$M_{t+1}^{\text{LO}} = b_{23} M_t^{\text{UO}} + \left(1 - b_{23} \frac{M_{\text{EQ}}^{\text{UO}}}{M_{\text{EQ}}^{\text{LO}}}\right) M_t^{\text{LO}} \quad (\nu_t^{\text{LO}})$$

$$T_{t+1}^{\text{AT}} = T_t^{\text{AT}} + c_1 \left(F_{2\text{XCO}_2} \frac{\log(M_t^{\text{AT}}/M_{\text{base}}^{\text{AT}})}{\log(2)} + F_t^{\text{EX}} \right) - c_1 \frac{F_{2\text{XCO}_2}}{T_{2\text{XCO}_2}} T_t^{\text{AT}} - c_1 c_3 (T_t^{\text{AT}} - T_t^{\text{OC}}) \quad (\eta_t^{\text{AT}})$$

$$T_{t+1}^{\text{OC}} = T_t^{\text{OC}} + c_4 (T_t^{\text{AT}} - T_t^{\text{OC}}) \quad (\eta_t^{\text{OC}})$$

$$0 \leq \mu_t \leq 1 \quad (\lambda_t^\mu)$$

One economic constraint, **five climate constraints**, one inequality constraint — the Lagrange/KKT multipliers in parentheses will all reappear as **network outputs** in Part III.

The SCC and the Optimal Carbon Tax in DICE

The SCC is the planner's marginal rate of substitution between atmospheric carbon and capital:

$$\text{SCC}_t = - \frac{\partial V_t / \partial M_{\text{AT},t}}{\partial V_t / \partial K_t}$$

In the optimum, an interior abatement policy $0 < \mu_t < 1$ equates the SCC with the marginal abatement cost, which defines the optimal carbon tax:

$$\text{CT}_t = \frac{\theta_{1,t} \theta_2 \mu_t^{\theta_2-1}}{\sigma_t}.$$

Preview of Part III

$\partial V_t / \partial M_{\text{AT},t}$ and $\partial V_t / \partial K_t$ are exactly the **shadow prices** ν_t^{AT} and λ_t of the planner problem. The DEQN learns both as network outputs — so the SCC comes **for free** with the solution, no numerical differentiation of a value function required.

Part III

Solving DICE with Deep Equilibrium Nets

DEQN in a Nutshell (Recap from Lecture 1)

Approximate the unknown policy $p : X \rightarrow Y \subset \mathbb{R}^K$ of a model characterized by first-order equilibrium conditions $G(\mathbf{x}, p(\mathbf{x})) = 0$ for all $\mathbf{x} \in X$ with a neural network $p(\mathbf{x}) \approx \mathcal{N}_\varphi(\mathbf{x})$, and use the residual of those conditions as the loss:

$$\ell_\varphi := \frac{1}{N_{\text{path length}}} \sum_{\mathbf{x}_t \text{ on sim. path}} \sum_{m=1}^{N_{\text{eq}}} (G_m(\mathbf{x}_t, \mathcal{N}_\varphi(\mathbf{x}_t)))^2, \quad \varphi'_k = \varphi_k - \alpha^{\text{learn}} \frac{\partial \ell(\varphi)}{\partial \varphi_k}.$$

Four building blocks:

1. A deep neural network approximating the equilibrium policies (Xavier/Glorot init);
2. A loss function: the error in the equilibrium conditions;
3. An updating mechanism: gradient descent (Adam);
4. A sampling method: states visited by **simulating the model** under the current policy.

Notation: we write φ for the network weights (Lecture 1 used ρ , which here denotes the rate of time preference).

What Breaks When We Point This at DICE?

In Lecture 1, every model (Brock–Mirman, stochastic growth, IRBC) was **stationary**: a time-invariant policy on an ergodic state space. DICE is not.

Three obstacles

1. **Non-stationarity**: exogenous trends make the policy depend on calendar time.
2. **Unbounded inputs and levels**: $t \in [0, \infty)$; K_t, C_t grow without bound along the trend.
3. **Occasionally binding constraint**: $\mu_t \leq 1$ once the backstop becomes cheap.

Three fixes

1. Add **time as an exogenous state**.
2. Compress time (**Traeger transform**) and **detrend** to efficiency units.
3. Smooth KKT via **Fischer–Burmeister** — the same trick as IRBC irreversibility.

And one genuinely new feature: the network must also output **shadow prices of the climate states**.

Fix 1: Time as an Exogenous State Variable

The state of the economy at time t :

$$\mathbf{x}_t := \left(k_t, M_t^{\text{AT}}, M_t^{\text{UO}}, M_t^{\text{LO}}, T_t^{\text{AT}}, T_t^{\text{OC}}, t \right)^T \in \mathbb{R}^7.$$

- ▶ Six **endogenous** states: detrended capital k_t , three carbon masses, two temperatures.
- ▶ One **exogenous** state: physical time t , which indexes all the deterministic trends $A_t, L_t, \sigma_t, \theta_{1,t}, E_{\text{Land},t}, F_t^{\text{EX}}$.

Interpretation

Feeding t to the network restores a **recursive** formulation: the policy is again a time-invariant function — of an enlarged state that includes the calendar.

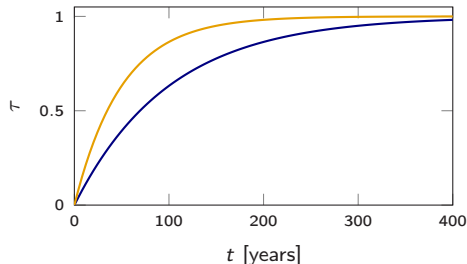
Dimensionality here comes from the **climate block**, not from many countries as in the IRBC.

Fix 2a: Compressing the Infinite Horizon (Traeger 2014)

Physical time is unbounded, $t \in [0, \infty)$ — a poor input for a neural network. Map it into the unit interval via the strictly monotonic transformation

$$\tau = 1 - \exp(-\vartheta t), \quad \tau \in (0, 1]$$

with inverse $t = -\frac{\ln(1 - \tau)}{\vartheta}$.



- ϑ controls where resolution is spent: small ϑ stretches the transition, large ϑ compresses it.
- The far future — where the detrended model settles down — collapses into a neighborhood of $\tau = 1$.
- Alternative for transition paths: shooting methods; the recursive

Fix 2b: Detrending to Efficiency Units

Levels grow without bound along the trend, so re-scale by effective labor $A_t L_t$:

$$c_t := \frac{C_t}{A_t L_t}, \quad k_t := \frac{K_t}{A_t L_t},$$

and normalize *all* Lagrange/KKT multipliers by the marginal-utility trend:

$$\hat{\lambda}_t := \frac{\lambda_t}{A_t^{1-\frac{1}{\psi}} L_t}, \quad \text{and likewise } \hat{\lambda}_t^\mu, \hat{v}_t^{\text{AT}}, \hat{v}_t^{\text{UO}}, \hat{v}_t^{\text{LO}}, \hat{\eta}_t^{\text{AT}}, \hat{\eta}_t^{\text{OC}}.$$

Growth is absorbed into an **effective discount factor**, which is now time-dependent:

$$\hat{\beta}_t := \exp \left(-\rho + \left(1 - \frac{1}{\psi} \right) g_t^A + g_t^L \right)$$

Why bother?

Networks train poorly on inputs and outputs whose scale drifts over the simulation. After detrending, all quantities stay $\mathcal{O}(1)$ along the path — recall the loss-normalization lessons from Lecture 1.

The Policy Network: 9 Outputs

The function approximated by the network maps the 7-dimensional state to

$$\mathcal{N}_\varphi(\mathbf{x}_t) \in \mathbb{R}^9 := \left(k_{t+1}, \mu_t, \hat{\lambda}_t, \hat{\lambda}_t^\mu, \hat{\nu}_t^{\text{AT}}, \hat{\nu}_t^{\text{UO}}, \hat{\nu}_t^{\text{LO}}, \hat{\eta}_t^{\text{AT}}, \hat{\eta}_t^{\text{OC}} \right).$$

2 choice variables

- ▶ k_{t+1} : next-period capital (used instead of c_t — simpler state update; the FOCs w.r.t. both are kept as **redundant information**, which helps training)
- ▶ μ_t : abatement rate

7 multipliers

- ▶ $\hat{\lambda}_t$: budget constraint
- ▶ $\hat{\lambda}_t^\mu$: KKT multiplier on $\mu_t \leq 1$
- ▶ $\hat{\nu}_t^{\text{AT/UO/LO}}$: carbon-cycle equations
- ▶ $\hat{\eta}_t^{\text{AT/OC}}$: temperature equations

Compare Lecture 1: the IRBC network output was $(k^{1'}, \dots, k^{N'}, \lambda, \mu^1, \dots, \mu^N)$ — multipliers only on *economic* constraints. Here, five of the nine outputs are shadow prices of **physical** equations.

Why Learn Shadow Prices for Non-Choice States?

A distinctive feature of IAM first-order conditions

FOCs are needed not only w.r.t. the economic choice variables (c_t, μ_t), but also w.r.t. the **climate variables**, even though the planner does not choose them.

- ▶ A marginal unit of emissions today changes M^{AT} , which redistributes through the ocean reservoirs, alters radiative forcing, warms both temperature layers, and only *then* feeds back into output via $\Omega(T^{\text{AT}})$ — over **centuries**.
- ▶ The marginal welfare effect of this chain cannot be computed analytically.
- ▶ Attaching a multiplier to *every* climate equation lets the network learn the **shadow price of each physical constraint**; the FOCs link them recursively across the chain.

Payoff

$\text{SCC}_t \propto -\hat{v}_t^{\text{AT}}/\hat{\lambda}_t$ — the headline object of climate economics is a ratio of two network outputs.

The Lagrangian (Detrended)

$$\begin{aligned} \mathcal{L} = \sum_{t=0}^{\infty} \hat{\beta}_t & \left[\frac{c_t^{1-1/\psi} - A_t^{1/\psi-1}}{1-1/\psi} + \hat{\lambda}_t \left\{ (1 - \Omega(T_{AT,t}) - \Theta(\mu_t)) k_t^\alpha + (1 - \delta) k_t - c_t - e^{g_t^A + g_t^L} k_{t+1} \right\} + \hat{\lambda}_t^\mu \{1 - \mu_t\} \right. \\ & + \hat{\nu}_t^{AT} \left\{ (1 - b_{12}) M_t^{AT} + b_{12} \frac{M_{EQ}^{AT}}{M_{EQ}^{UO}} M_t^{UO} + \sigma_t (1 - \mu_t) A_t L_t k_t^\alpha + E_t^{Land} - M_{t+1}^{AT} \right\} + \hat{\nu}_t^{UO} \{\dots\} + \hat{\nu}_t^{LO} \{\dots\} \\ & \left. + \hat{\eta}_t^{AT} \left\{ T_t^{AT} + c_1 \left(F_{2XCO2} \frac{\log(M_t^{AT}/M_{base}^{AT})}{\log(2)} + F_t^{EX} \right) - c_1 \frac{F_{2XCO2}}{T_{2XCO2}} T_t^{AT} - c_1 c_3 \left(T_t^{AT} - T_t^{OC} \right) - T_{t+1}^{AT} \right\} + \hat{\eta}_t^{OC} \{\dots\} \right] \end{aligned}$$

- Each $\{\cdot\}$ is one constraint from the planner problem, in detrended units; $\hat{\nu}^{UO}, \hat{\nu}^{LO}, \hat{\eta}^{OC}$ rows omitted for space (structure identical to their AT counterparts).
- Differentiating w.r.t. c_t, μ_t, k_{t+1} , and each of the five climate states at $t + 1$ yields the FOCs on the next slides.
- The consumption FOC, $c_t^{-1/\psi} A_t^{1/\psi-1} L_t = \hat{\lambda}_t$, is used to substitute c_t out of the budget constraint.

Fix 3: The Backstop Bound via Fischer–Burmeister

The abatement FOC defines the KKT multiplier on $\mu_t \leq 1$:

$$\hat{\lambda}_t^\mu := -\hat{\lambda}_t \Theta'(\mu_t) k_t^\alpha - \hat{v}_t^{\text{AT}} \sigma_t A_t L_t k_t^\alpha, \quad \text{with} \quad 1 - \mu_t \geq 0 \perp \lambda_t^\mu \geq 0.$$

As in Lecture 1 (IRBC irreversibility), replace the complementarity condition with the smooth **Fischer–Burmeister** function:

$$\psi^{\text{FB}}(\hat{\lambda}_t^\mu, 1 - \mu_t) = \hat{\lambda}_t^\mu + (1 - \mu_t) - \sqrt{(\hat{\lambda}_t^\mu)^2 + (1 - \mu_t)^2} = 0$$

Economics of the corner

- ▶ Early on: backstop expensive, $\mu_t < 1$ interior, $\lambda_t^\mu = 0$.
- ▶ Eventually $\theta_{1,t} \downarrow$: full abatement $\mu_t = 1$ binds, $\lambda_t^\mu > 0$.

Same trick, new constraint

- ▶ IRBC: $\text{FB}(\mu^j, I^j)$ for irreversible investment.
- ▶ DICE: $\text{FB}(\hat{\lambda}^\mu, 1 - \mu)$ for the backstop bound.

The Loss Function I: Economic Residuals

Three of the eight loss components involve the economic block:

$$l_1 := e^{g_t^A + g_t^L} \hat{\lambda}_t - \hat{\beta}_t \left[\hat{\lambda}_{t+1} \left((1 - \Omega(T_{AT,t+1}) - \Theta(\mu_t)) \alpha k_{t+1}^{\alpha-1} + (1 - \delta) \right) + \hat{\nu}_{t+1}^{AT} (1 - \mu_{t+1}) \sigma_{t+1} A_{t+1} L_{t+1} \alpha k_{t+1}^{\alpha-1} \right] \quad (\text{Euler equation for capital})$$

$$l_2 := (1 - \Omega(T_{AT,t}) - \Theta(\mu_t)) k_t^\alpha + (1 - \delta) k_t - c_t - e^{g_t^A + g_t^L} k_{t+1} \quad (\text{budget constraint})$$

$$l_8 := \hat{\lambda}_t^\mu + (1 - \mu_t) - \sqrt{\left(\hat{\lambda}_t^\mu \right)^2 + (1 - \mu_t)^2} \quad (\text{Fischer–Burmeister})$$

- ▶ l_1 is the familiar capital Euler equation — **plus a new term**: investing more capital raises output, hence emissions, hence future climate damage, priced by $\hat{\nu}_{t+1}^{AT}$.
- ▶ In l_2 , consumption is substituted via the FOC $c_t = \left(\hat{\lambda}_t / (A_t^{1-1/\psi} L_t) \right)^{-\psi}$.

The Loss Function II: Climate Shadow-Price Recursions

The remaining five components propagate marginal effects through the climate chain:

$$l_3 := \hat{v}_t^{\text{AT}} - \hat{\beta}_t \left[\hat{v}_{t+1}^{\text{AT}} (1 - b_{12}) + \hat{v}_{t+1}^{\text{UO}} b_{12} + \hat{\eta}_{t+1}^{\text{AT}} c_1 F_{2\text{XCO}_2} \frac{1}{\ln 2 M_{\text{AT},t+1}} \right] \quad (\text{carbon AT})$$

$$l_4 := \hat{v}_t^{\text{UO}} - \hat{\beta}_t \left[\hat{v}_{t+1}^{\text{AT}} b_{12} \frac{M_{\text{EQ}}^{\text{AT}}}{M_{\text{EQ}}^{\text{UO}}} + \hat{v}_{t+1}^{\text{UO}} \left(1 - b_{12} \frac{M_{\text{EQ}}^{\text{AT}}}{M_{\text{EQ}}^{\text{UO}}} - b_{23} \right) + \hat{v}_{t+1}^{\text{LO}} b_{23} \right] \quad (\text{carbon UO})$$

$$l_5 := \hat{v}_t^{\text{LO}} - \hat{\beta}_t \left[\hat{v}_{t+1}^{\text{UO}} b_{23} \frac{M_{\text{EQ}}^{\text{UO}}}{M_{\text{EQ}}^{\text{LO}}} + \hat{v}_{t+1}^{\text{LO}} \left(1 - b_{23} \frac{M_{\text{EQ}}^{\text{UO}}}{M_{\text{EQ}}^{\text{LO}}} \right) \right] \quad (\text{carbon LO})$$

$$l_6 := \hat{\eta}_t^{\text{AT}} - \hat{\beta}_t \left[-\hat{\lambda}_{t+1} \Omega'(T_{\text{AT},t+1}) k_{t+1}^\alpha + \hat{\eta}_{t+1}^{\text{AT}} \left(1 - c_1 \frac{F_{2\text{XCO}_2}}{T_{2\text{XCO}_2}} - c_1 c_3 \right) + \hat{\eta}_{t+1}^{\text{OC}} c_4 \right] \quad (\text{temperature AT})$$

$$l_7 := \hat{\eta}_t^{\text{OC}} - \hat{\beta}_t \left[\hat{\eta}_{t+1}^{\text{AT}} c_1 c_3 + \hat{\eta}_{t+1}^{\text{OC}} (1 - c_4) \right] \quad (\text{temperature OC})$$

Total loss, evaluated along the simulated path (cf. Lecture 1):

$$\ell_\varphi := \frac{1}{N_{\text{path length}}} \sum_{\mathbf{x}_t \text{ on sim. path}} \sum_{m=1}^{N_{\text{eq}}=8} (l_m(\mathbf{x}_t, \mathcal{N}_\varphi(\mathbf{x}_t)))^2$$

Reading the Shadow-Price Recursions

The five climate residuals l_3 – l_7 have a common structure — each is a **backward recursion** for a shadow price, mirroring the forward transition of its state:

$$\underbrace{\hat{\nu}_t}_{\text{price today}} = \hat{\beta}_t \left[\underbrace{(\text{persistence}) \times \hat{\nu}_{t+1}}_{\text{carbon stays put}} + \underbrace{(\text{transfer}) \times \hat{\nu}_{t+1}^{\text{other box}}}_{\text{carbon moves on}} + \underbrace{(\text{forcing/damage term})}_{\text{carbon does harm}} \right]$$

- ▶ l_3 : atmospheric carbon either stays $(1 - b_{12})$, moves to the upper ocean (b_{12}) , or raises forcing — priced by $\hat{\eta}_{t+1}^{\text{AT}} c_1 F_{2\text{XCO}_2} / (\ln 2 M_{\text{AT},t+1})$.
- ▶ l_6 : atmospheric warmth damages output *today* $(-\hat{\lambda}_{t+1} \Omega' k_{t+1}^\alpha)$, persists, or leaks into the deep ocean.
- ▶ l_7 : deep-ocean warmth is a *pure storage* term — it only matters because it flows back to the atmosphere $(c_1 c_3)$.

The economic content

This chain is precisely how one marginal ton of carbon today is converted into a discounted stream of future consumption losses — i.e., how the model computes the SCC.

Simulating the Path and Training the Network

Algorithm: DEQN training for CDICE

- 1: Initialize φ (Xavier/Glorot)
- 2: **repeat**
- 3: From $\mathbf{x}_0$ (2015 calibration), simulate $N_{\text{path length}}$ states:
- 4: k_{t+1} from the network output
- 5: climate states from their laws of motion
- 6: $t \leftarrow t + 1$
- 7: Evaluate ℓ_φ (8 squared residuals, path average)
- 8: Adam update: $\varphi \leftarrow \varphi - \alpha^{\text{learn}} \nabla_\varphi \ell$
- 9: **until** $\ell_\varphi < \epsilon$

State transition

$$\mathbf{x}_{t+1} = (k_{t+1}, M_{t+1}^{\text{AT}}, \dots, T_{t+1}^{\text{OC}}, t+1)^T$$

- **No expectation operator**: the model is deterministic, so no quadrature nodes (Gauss–Hermite, monomial) as in Lecture 1 — the residuals are evaluated on the single trajectory.
- The path starts from the **2015 initial condition**, not from an ergodic distribution: we solve a transition path.

Architecture and hyperparameters (layers, width, α^{learn} , $N_{\text{path length}}$, ϑ) live in the replication repo:

https://github.com/ClimateChangeEcon/Climate_in_Climate_Economics.

Accuracy: Error Statistics at Convergence

Summary statistics of the loss components along the simulated path, benchmark CDICE:

	l_1	l_2	l_3	l_4	l_5	l_6	l_7	l_8
mean	1.1e-4	5.3e-4	2.3e-4	3.8e-4	1.2e-4	1.8e-5	3.1e-5	5.0e-4
std	6.4e-5	1.2e-4	1.2e-4	7.8e-5	6.3e-5	1.5e-5	2.4e-5	3.2e-4
median	1.2e-4	5.3e-4	2.6e-4	3.6e-4	9.7e-5	1.3e-5	2.6e-5	5.1e-4
max	3.3e-4	9.8e-4	5.4e-4	7.4e-4	3.1e-4	8.5e-5	1.3e-4	1.3e-3

- ▶ Following the literature (Judd et al.), the **maximum error** is defined as the 99.9% quantile of the error distribution.
- ▶ Mean residuals of order 10^{-4} , maxima of order 10^{-3} — comparable to the Lecture 1 IRBC results; all model variants in the paper reach a similar accuracy.

Part IV

What Is Different Compared to Lecture 1?

From Brock–Mirman and IRBC to DICE

	Brock–Mirman	IRBC	DICE/CDICE
States	(K, z) : 2	$2N$	7 (incl. time τ)
Stochastic	yes (1 shock)	yes ($N + 1$ shocks)	no (deterministic)
Stationary	yes	yes	no (exogenous trends)
Solution object	ergodic policy	ergodic policy	transition path from 2015
Network outputs	1	$2N + 1$	9 (2 choices + 7 multipliers)
Loss components	1 Euler	N Euler + ARC + N FB	8 (Euler, budget, 5 shadow price)
Expectations	quadrature	monomial rule	none
Binding constraint	none	irreversibility (FB)	backstop $\mu \leq 1$ (FB)
Detrending	not needed	not needed	efficiency units + $\hat{\beta}_t$

Same DEQN recipe throughout — what changes is the structure of the state space and of the loss, not the method.

Five Things That Are Genuinely New Here

1. **Deterministic transition path.** No shocks, no expectation operator, no quadrature: the loss is evaluated along one trajectory from the 2015 state, not on an ergodic cloud.
2. **Non-stationarity.** Time enters as an exogenous state; the Traeger transform $\tau = 1 - e^{-\vartheta t}$ makes it a bounded network input.
3. **Detrending.** Capital, consumption, and all multipliers are rescaled by trends in A_t, L_t ; growth is absorbed into the effective discount factor $\hat{\beta}_t$.
4. **Shadow prices of physical states.** Five of nine network outputs price climate equations the planner never chooses — and deliver $\text{SCC}_t \propto -\hat{v}_t^{\text{AT}} / \hat{\lambda}_t$ for free.
5. **Dimensionality from physics.** The state grows because of the climate emulator (3 carbon + 2 temperature states), not because of more agents or countries.

And two things that are exactly the same

Residual-based loss on simulated states, and the Fischer–Burmeister treatment of occasionally binding constraints — both carried over from Lecture 1 unchanged.

Recap: Three Building Blocks

Model

CDICE: Ramsey growth + 3-box carbon cycle + 2-layer temperature, recalibrated to CMIP5 benchmarks; non-stationary via 6 exogenous trends.

Network

$\mathcal{N}_\varphi : \mathbb{R}^7 \rightarrow \mathbb{R}^9$ on the detrended state (with time τ); outputs 2 choices and 7 normalized shadow prices.

Training

Simulate the deterministic path from 2015, Adam on the sum of 8 squared FOC residuals; mean errors $\sim 10^{-4}$.

$\Rightarrow$ **The SCC and the optimal carbon tax are read directly off the converged network.**

Bridge: From the Deterministic Baseline Onwards

What we leave with today. A non-stationary IAM solved with the Lecture 1 DEQN recipe plus three fixes (time as a state, Traeger compression, detrending) and one new idea (learned shadow prices for physical constraints).

Natural extensions.

- ▶ *Stochastic* IAMs: TFP and climate-sensitivity risk, Epstein–Zin preferences — expectations (and quadrature) return.
- ▶ *Pseudo-states*: add uncertain parameters (ECS, damage curvature, ρ , ψ) as inputs and solve a **family** of models once.

Closing the loop with Part 1.

- ▶ GP surrogates map uncertain parameters to the SCC; Bayesian active learning picks where to solve next.
- ▶ Result: SCC **distributions** and global sensitivity indices instead of point estimates.

References. Folini, Friedl, Kübler & Scheidegger (2025, *REStud*); Azinovic, Gaegauf & Scheidegger (2022, *IER*); Traeger (2014); Nordhaus (2008, 2017).

Questions?

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<https://sischei.github.io>

Materials: https://github.com/sischei/cemracs2026_DEQN

Next up:

Hands-on: solving CDICE with the
production DEQN code

[https:
//github.com/ClimateChangeEcon/
Climate_in_Climate_Economics](https://github.com/ClimateChangeEcon/Climate_in_Climate_Economics)